

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
LONERE – RAIGAD -402 103
Supplementary Winter Examination–2023**

Course: B. Tech

Sem.: VI

Branch: Electronics and Communication Engineering / Electronics and Telecommunication Engineering

Subject with Subject Code: - Digital Image Processing (BTETC603)

Marks: 60

Date: 20/01/2024

Time: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	CO	Marks
Q. 1 Solve Any Two of the following.		12
A) Define Image, Brightness, Dynamic Range, Gray level, Hue, Saturation.	CO1	6
B) Draw and explain the fundamental steps in digital image processing.	CO1	6
C) Define 2D convolution. Write down the properties of 2D convolution.	CO1	6
Q.2 Solve Any Two of the following.		12
A) Write a short note on a) Match Band Effect b) Simultaneous contrast point	CO3	6
B) Write a short note on a) Spatial effects in color vision b) Optical illusions.	CO3	6
C) Define and explain the types of neighbours and connectivity in image processing.	CO1	6
Q. 3 Solve Any Two of the following.		12
A) Explain the concepts of a) Non rectangular Grid sampling b) Hexagonal sampling	CO1	6
B) Derive the reconstruction level equation for optimum Mean Square Lloyd-Max quantizer.	CO4	6
C) Explain the concept of vector quantization in image processing.	CO1	6

Q.4 Solve Any Two of the following. **12**

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| A) Define DCT along with forward and backward transformation. What is its need? | CO2 | 6 |
| B) Explain all steps perform to get transformed matrix using K-L transformation. | CO1 | 6 |
| C) Write notes on Walsh Transform and Hadamard Transform for 1D and 2D. | CO1 | 6 |

Q. 5 Solve Any Two of the following. **12**

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| A) Define histogram. What is the need of histogram equalization? Write the steps for histogram equalization. | CO4 | 6 |
| B) Write a note on Roberts, Prewitt and Sobel operator for edge detection in an image. | CO2 | 6 |
| C) What is meant by Image Restoration? Draw the degradation model and explain. | CO4 | 6 |

***** End *****